

The Effectiveness of Performance Signals in Formula One Driver Promotion: Evidence from Pre- and Post-Formalization Junior Series

Dhairya Patankar (1009111934) *

Department of Economics, University of Toronto

ECO420

Professor Nicholas Zammit

April 28, 2026

Abstract

It is well-documented that performance outcomes play a central role in determining advancement in competitive labor markets. This study examines whether the effectiveness of early-career performance signals in predicting promotion to Formula One has changed following the formalization of the junior formula system in 2017. I employ a cross-sectional framework using data on drivers competing in Formula 2 (F2) and GP2 from 2008 to 2025, where promotion is defined as entry into Formula One within three years of a driver's final junior season. Performance is captured using a composite index constructed from standardized measures of race results, qualifying performance, and reliability. The impact is estimated using a probit regression model with clustered standard errors, controlling for team quality and time effects. The results indicate that performance is a strong and statistically significant predictor of promotion; however, there is no evidence that its effectiveness has changed following formalization.

*Contact email: dhairya.patankar@mail.utoronto.ca I would like to thank Professor Nicholas Zammit for his guidance and support throughout this project.

1 Introduction

At the end of the 2023 season and heading into 2024, Formula One (F1) experienced an unprecedented outcome: for the first time in its history, no teams changed their driver lineups. In contrast, the transition into 2025 saw substantial turnover, with five rookie drivers entering F1, only two of whom had finished in the top five of the 2024 Formula 2 (F2) standings. At the same time, recent F2 champions Felipe Drugovich and Theo Pourchaire remain without F1 seats despite strong feeder-series performance. This variation suggests that promotion to F1 does not respond mechanically to junior results. This paper examines whether the effectiveness of early-career performance signals has changed over time, comparing the pre-formalization era (2008–2016) to the modern F2/F3 system.

This question is of broader economic interest because it relates to how signals function in structured labour markets. In theory, the formalization of feeder series, through standardized regulations and clearer progression pathways, should improve the informational content of performance metrics. Junior results would then more accurately reflect driver ability and serve as stronger signals for promotion. However, persistent inconsistencies suggest that formalization may not have strengthened signal effectiveness. This is particularly relevant in an increasingly saturated driver market, where the number of qualified candidates has expanded while the number of F1 seats remains fixed.

The institutional structure of F1 helps explain this dynamic. Müller-Kock and Budzinski (2016) show that the sport operates under a centralized revenue system that provides teams with stable income, weakly tied to marginal performance differences. This reduces incentives to take risks on new drivers and may reinforce incumbent advantages. In such an environment, even higher-quality performance signals may not translate into higher promotion probabilities. This paper contributes to the literature on signalling and labour market selection by testing whether formalization strengthens the predictive power of performance metrics. To address this question, I construct a driver-level dataset spanning both the pre-formalization era (2008–2016) and the post-formalization F2/F3 system (2017–present). I aggregate performance into a composite index combining championship position, points finishing rate, and pole position rate, while accounting for unreliability (DNF/DSQ/DNS rate). I also control for team quality using constructors’ championship position. Using a probit framework, I estimate the relationship between this index and the probability of promotion to Formula One.

The results show that the performance index is a strong and statistically significant predictor of promotion. However, there is no evidence that its effectiveness has improved in the post-formalization era, as the interaction between performance and the formalized regime is small and statistically insignificant. Team performance also plays a significant role. Notably, the average finishing position of promoted drivers is higher in the post-formalization era, indicating an increase in entrant quality. Taken together, these findings suggest that while talent has risen, the marginal effectiveness of performance signals has not, reinforcing the view that the F1 driver market remains shaped by structural constraints rather than

purely by merit.

2 Literature Review

Competing at the highest level in any sport implies a cutthroat environment driven by results. Professional sport labour markets are characterized by elite talent, large rewards for those few who win, and an extreme scarcity of opportunities. These features contribute to an oversupply of drivers, as there are so many young talents who dream of competing at the highest stage of their respective sports, despite there being very limited opportunities to compete. Formula One (F1) represents an extreme example of such a labour market, with only twenty seats spread across ten teams and large pipeline of junior drivers, all competing for entry. In this context, junior series and rookie season performances serve as signals of a driver’s abilities. However, incumbent advantages, institutional constraints, and high switching costs present challenges in translating performance into promotion opportunities. This section explores the existing literature regarding oversaturation in sport labour markets, persistence of incumbent advantages, and the role of signaling in determining promotion.

Unlike conventional labour markets, oversupply in professional sports is not viewed as a market failure but as a rational consequence of highly skewed rewards, rank-based competitive incentives, and fixed demand. Rosen and Sanderson (2000) emphasize the role of fixed demand in sports labour markets, demonstrating how many aspiring athletes compete for a small number of roster slots. For example, they note that, “In the U.S. perhaps, 30 new players are talented enough to make it into [the] 320 player roster of the NBA in a year, and those 30 players started out as more than 10,000 high school seniors who then became less than 1,000 freshman players at Division I universities.” This example highlights that even for the most talented players, reaching the highest level of a sport is highly unlikely, in part because the number of available slots is strictly limited. Despite this low probability of reaching the highest level, excess supply still occurs in equilibrium, suggesting that entry decisions are not only driven by expected success but are also tied to the presence of highly skewed earnings for the top performers.

Both Rosen and Sanderson (2000) and Scarfe et al. (2021) address the highly skewed earnings distributions observed in professional sports labour markets, with the latter paper explicitly characterizing the pattern as the ‘superstar effect’. The former paper notes that earnings in professional sports are highly skewed, with a small number of athletes capturing a disproportionate share of the league’s total compensation. Building on this note, Scarfe et al. (2021) argue that the extreme upper-tail earnings are driven not only by inherent talent but also fame and visibility. To demonstrate this, they decompose Major League Soccer superstars’ wages into a predicted component, explained by past on-field performance, and a residual component correlated with popularity and off-field value. They find that high wages at the top are not driven entirely by performance, but by popularity, as superstar players can increase revenue without improving team performance.

The combination of fixed roster slots and skewed payoffs presents incentives to aspiring athletes to prove they are better relative to their rivals, promoting rank order competition. Rosen and Sanderson (2000) demonstrate this competition through an “arms race” analogy, in which opportunities are allocated based on relative results rather than absolute performance. In this oversaturated labour market, the question shifts from why there are so many entrants to who teams will select and why incumbent athletes often retain their positions despite strong performances from potential entrants.

Existing literature argues that incumbent advantages play a central role in teams’ selection of talent, allowing experienced players to retain positions despite performing comparably to, or worse than potential entrants. Peeters et al. (2022) document this mechanism by showing teams may retain mediocre players when uncertainty regarding entrants is high enough. This retention of mediocre players is driven by asymmetric information, as teams face have more information regarding the ability of the incumbent than the entrant. As a result, teams may be more hesitant to hire new talent and instead prefer retaining the incumbent, which minimizes their risk. For example, even if the expected performance between the incumbent and entrant is similar, the entrant’s skill has a higher variance as they do not have the experience of competing at the top level yet, and this high variance is part of what drives teams away from hiring new talent. The incumbents’ experience also proves invaluable in retaining their position, as it works as information capital. As entrants lack this capital, it can lead to low turnover in the labour market and continuous occupancies of elite positions. These barriers to entry are most prevalent in markets that have a fixed demand, high switching costs, and a long learning process.

As established, in these markets, firms face asymmetric information regarding incumbents, requiring entrants to rely on performance signals as a substitute for the information capital they lack. However, strong incumbent advantages weaken the effectiveness of these signals, as strong early performance signals may fail in overcoming entrants’ informational disadvantages. This dynamic is explored by Arcidiacono et al. (2010), who develop an employer-learning framework in which worker ability is initially unobserved by firms and gradually learned through performance. In this framework, these early performances constitute noisy signals of latent ability, which evolve as entrants gain more experience. Because these signals are an imprecise reflection of ability, in markets with large consequences associated with incorrect promotion, employers will rationally choose to discount these signals. As a result, Arcidiacono et al. (2010) implies that under high uncertainty, the effectiveness of entrant performance signals erode in promotion decisions.

Taken together, fixed demand, highly skewed earnings, incumbent advantages, and signal erosion motivate modeling promotion as a binary outcome that can be analyzed using a probabilistic model. Because promotion in F1 is binary, we use a latent variable framework in which promotion depends on observable performance signals and unobserved ability. This latent variable structure is consistent with the employer-learning framework in Arcidiacono et al. (2010), where worker ability is imprecisely observed and revealed through performance signals. In this framework, employer decisions are driven by whether the expected abil-

ity, conditional on observed performance, is above a certain threshold, making promotion expressible as a discrete outcome.

Following Altonji and Pierret (2001), who model discrete employment outcomes using a probit model, this paper estimates the probability of promotion as a function of observable performance signals and past results. In their paper, employers rely on observable characteristics of the entrants when uncertainty regarding their ability is high, making probit a preferable tool for capturing promotion decisions under uncertainty. The labour markets examined in both papers share key characteristics such as asymmetric information, noisy early performance signals, and high-pressure promotion decisions, which all closely align with the F1 driver market.

Taken together, the literature highlights how fixed demand, asymmetric information, and incumbent advantages shape promotion outcomes in highly competitive labour markets such as F1. While performance signals serve as key indicators of entrant ability, their effectiveness may be limited by informational frictions and structural constraints. These insights motivate the empirical framework of this paper, which models promotion as a probabilistic outcome driven by observable performance and institutional conditions. By testing whether the effectiveness of performance signals has changed over time, this analysis contributes to the literature by providing evidence on how these theoretical mechanisms operate in the context of the F1 driver market.

3 Data

3.1 Data Sources and Structure

The analysis uses an unbalanced panel dataset of junior drivers competing in Formula 2 (F2), GP2, Formula 3 (F3), and GP3 over the period 2008–2025. The unit of observation is at the driver-season level, where each observation captures a driver’s performance and competitive environment in a given season.

The data is compiled from official championship standings for F2 and F3, supplemented by archival records from Wikipedia for F2, F3, GP2, and GP3. These sources provide consistent information on performance outcomes, team affiliation, and race-level results, allowing for the construction of standardized performance measures across series and time. While minor discrepancies across sources may exist, the data is sufficiently consistent for comparative analysis.

This dataset is particularly well-suited to the research question, as junior formula series represent the primary pipeline into Formula One, with clearly observable performance metrics and well-defined promotion outcomes. The structured nature of these series allows for a consistent comparison of performance signals across drivers and over time.

3.2 Sample Construction

The primary sample includes all drivers who competed in F2 or GP2, tracking whether they are promoted to Formula One within three years of their final junior-season appearance. This time window reflects the lag in promotion decisions while limiting contamination from longer-term career dynamics.

Drivers are included if they participated in at least one F2/GP2 season, regardless of whether participation was full or partial. The final sample consists of 259 drivers and 540 observations. To examine changes over time, the dataset is split into a pre-formalization period (2008–2016) and a post-formalization period (2017–2025), reflecting the restructuring of feeder series under the FIA and the introduction of a more standardized development pathway.

As a comparative benchmark, a second dataset is constructed using GP3 and F3 data. In this setting, the dependent variable captures promotion to GP2/F2 within three years of a driver’s final season, using the same inclusion criteria as the primary sample. An additional specification allows for direct promotion from GP3 to F1, reflecting the five observed cases of such transitions. This sample includes 305 drivers and 480 observations and is divided into pre-formalization (2010–2016) and post-formalization (2019–2025) periods. The 2017–2018 seasons are excluded to maintain balanced time windows across regimes.

3.3 Key Variables Description

The dependent variable in the primary dataset, *Promoted*, is a binary indicator equal to one if a driver enters Formula One within three years of their final junior season, and zero otherwise. The main explanatory variable is the performance index (*PerfIndex*), a composite measure that aggregates standardized indicators of championship position (reversed so that higher values reflect stronger performance), points finishing rate, and pole position rate. To account for race-level uncertainty and reliability, the DNF/DSQ/DNS rate is included with a negative sign. All performance measures are expressed as rates by dividing raw values by the number of races each driver competed in, ensuring comparability across seasons. Higher values of the index therefore reflect stronger overall performance, capturing consistency, speed, and reliability in a single measure while reducing multicollinearity across individual metrics. As the performance index is constructed from standardized components, it is unitless and interpreted in terms of standard deviations from the mean, such that a one-unit increase corresponds to a one standard deviation improvement in performance.

Controls include team quality, proxied by constructors’ championship position (reversed), capturing differences in resources and competitiveness that may influence both observed performance and promotion outcomes, as well as a post-formalization indicator equal to one for seasons from 2017 onward. The secondary dataset follows the same structure, with dependent variables *PromotedF2* (promotion to GP2/F2) and *PromotedF1* (direct promotion to Formula One).

Table 1: Summary Statistics of Key Variables (F2/GP2 Sample)

Variable	Obs	Mean	Std. Dev.	Min	Max
Promoted	540	0.080	0.271	0	1
Performance Index	540	0.000	2.977	-8.069	11.576
Team Performance	540	-6.970	3.465	-13	-1

The descriptive statistics highlight several key patterns in the data. For the summary of all variables, please see A.2. The overall promotion rate in the main dataset is approximately 7.96%, reflecting the highly selective nature of progression to Formula One. When split by institutional period, the promotion rate is approximately 8.22% in the pre-formalization era (2008–2016) and 7.66% in the post-formalization era (2017–2025), indicating a slight decline over time. The performance index has a mean close to zero by construction and a standard deviation of approximately 2.98, indicating substantial variation in driver performance across the sample. This dispersion provides sufficient variation to identify differences in promotion probabilities across performance levels. Team quality, proxied by reversed constructors’ championship position, also exhibits meaningful variation across observations, capturing differences in competitive environments. Constructors’ championship position is reversed such that values closer to zero correspond to stronger team performance. The low overall promotion rate reflects a rare-event setting, which motivates the use of nonlinear models in the empirical analysis.

In addition to the main GP2/F2-to-F1 sample, the analysis incorporates promotion outcomes from lower tiers (F3/GP3 to GP2/F2) as a benchmark. These transitions occur in a less constrained environment and provide a useful comparison for assessing whether performance signals operate differently across levels. Summary statistics for this sample are reported in C.1. To provide a non-parametric illustration of the relationship between performance and promotion, Figures 1 and 2 present binned and grouped promotion rates across the performance distribution.

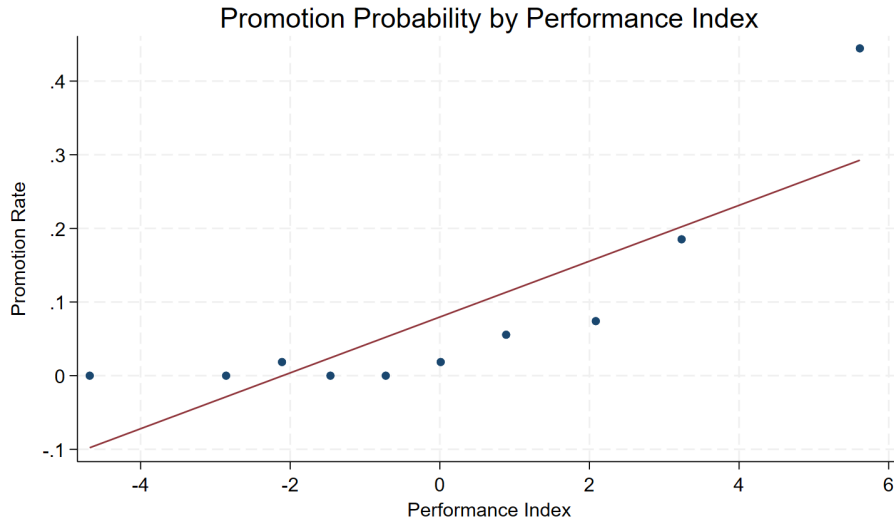


Figure 1: Promotion Probability by Performance Index

This figure shows a strong positive relationship between the performance index and the probability of promotion to Formula One. Promotion rates remain close to zero at lower levels of performance but increase steadily, and more sharply, among higher-performing drivers, suggesting that only top-performing drivers have meaningful chances of promotion.

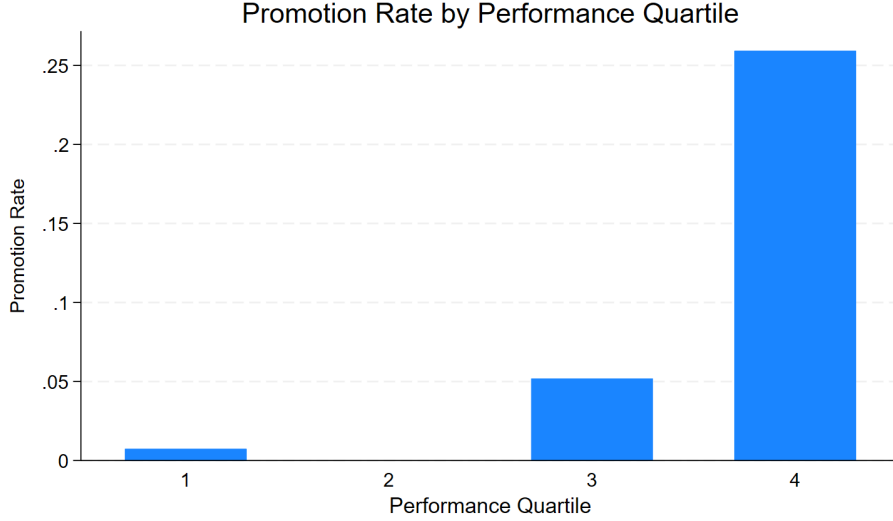


Figure 2: Promotion Rate by Performance Quartile

The quartile analysis reinforces this pattern, with promotion rates rising substantially across performance groups. Drivers in the top quartile experience significantly higher promotion rates compared to those in lower quartiles, while promotion is virtually nonexistent among the lowest-performing drivers, highlighting the highly selective nature of progression to Formula One.

These descriptive patterns provide initial evidence that performance plays a central role in promotion outcomes, a relationship that is examined more formally in the econometric analysis that follows.

4 Methodology

The empirical framework used in this analysis is a latent variable model estimated through a probit specification, which is appropriate given the binary nature of the outcome variable. The dependent variable captures whether a driver is promoted to Formula One within a defined time horizon. Let Y_i^* denote the latent propensity for driver i to obtain an F1 seat following their final junior-series season. The observed outcome Y_i is a binary indicator equal to one if the driver secures an F1 seat within three years of their final Formula 2 (or GP2)

season, and zero otherwise. Formally,

$$Y_i = \begin{cases} 1 & \text{if } Y_i^* > 0 \\ 0 & \text{otherwise} \end{cases}$$

This definition reflects the institutional reality that promotion decisions may not occur immediately, but instead within a short window following junior-series participation, while limiting contamination from longer-term career dynamics.

The baseline model is specified as:

$$Y_i^* = \alpha + \beta \text{PerfIndex}_i + \gamma \text{Post_Formalization}_i + \delta (\text{PerfIndex}_i \times \text{Post_Formalization}_i) + \theta \text{Controls}_i + \varepsilon_i$$

The central explanatory variable is a composite performance index, *PerfIndex_i*, designed to capture multiple dimensions of driver ability in a single measure. The index aggregates championship position (transformed such that higher values reflect better performance), points finishing rate, and pole position rate, while incorporating a control for race-level uncertainty through the driver’s DNF/DSQ/DNS rate. Each component is standardized prior to aggregation to ensure comparability across metrics. Because higher rates of non-finishes reflect poorer performance, this component enters the index with a negative sign. The resulting index therefore consistently increases with stronger overall performance, capturing a combination of speed, consistency, and reliability while mitigating multicollinearity across highly correlated indicators.

The vector *Controls_i* includes additional covariates that account for factors beyond individual performance. Team quality is proxied by the constructors’ championship position in each season, reflecting differences in equipment, resources, and competitive environments that may influence both observed outcomes and promotion probabilities. In addition, year fixed effects are included to account for time-specific factors, such as changes in grid availability, regulatory conditions, and broader market dynamics, that may influence promotion probabilities independently of individual performance. The indicator *Post_Formalization_i* identifies whether a driver competed in the post-2017 period, corresponding to the formalization of the F2 and F3 feeder system. The interaction between the performance index and this indicator allows the marginal effect of performance on promotion to vary across institutional regimes, providing a direct test of whether the effectiveness of performance signals has changed over time.

A linear probability model would be inappropriate in this context, as it does not constrain predicted probabilities to lie within the unit interval and imposes constant marginal effects across observations. The probit specification instead maps the latent index into probabilities through the cumulative normal distribution, allowing for nonlinear responses that better reflect the discrete and competitive nature of promotion decisions.

Earlier specifications included individual performance measures, such as championship position, win rates, and pole rates, entered separately. While informative, these models suffered from multicollinearity due to the high correlation among performance metrics, resulting in unstable coefficient estimates and reduced statistical precision. In addition, these specifications exhibited issues of separation, where certain combinations of predictors perfectly predicted promotion outcomes, leading to unreliable maximum likelihood estimates. The construction of the composite performance index addresses both concerns by consolidating information across metrics, reducing dimensionality, and stabilizing estimation within the probit framework. The results from the earlier specification can be found in B.1.

A key limitation of this approach is the potential for selection bias. Promotion to Formula One is not random, and unobserved factors, such as innate ability, financial backing, or institutional affiliations, may influence both measured performance and promotion outcomes. If these factors are correlated with the performance index, the estimated effect of performance may be biased upward. Conversely, measurement error in the index, particularly due to team-level influences on outcomes such as non-finishes, may attenuate the estimated relationship. While the inclusion of team-level controls partially mitigates these concerns, they cannot be fully eliminated within this framework.

To provide additional context, the analysis also considers promotion dynamics in lower tiers of the racing hierarchy, specifically transitions from GP3/F3 to GP2/F2. These lower-tier markets are less capacity-constrained and exhibit fewer institutional rigidities, making them a useful benchmark. If performance is found to be a stronger predictor of promotion at lower levels, this would suggest that institutional constraints at the F2-to-F1 transition weaken the role of performance signals. Conversely, similar effects across tiers would indicate that performance remains a consistent determinant of advancement throughout the racing ladder.

More advanced approaches, such as fixed effects or instrumental variable strategies, could in principle address unobserved heterogeneity more directly. However, given the limited panel dimension and the rarity of promotion events, these methods are difficult to implement without sacrificing statistical power or introducing additional identification challenges. As such, the probit framework represents a balanced approach that captures the key relationships of interest while maintaining interpretability and empirical tractability.

5 Results

This section presents empirical findings from the probit model estimating the relationship between driver performance and the probability of promotion to Formula One. The results focus on the role of the constructed performance index, the impact of institutional changes following formalization, and the robustness of these relationships across alternative specifications.

5.1 Baseline Results

Table 2: Results from F2/GP2 – F1 Model (Transposed)

	Perf. Index	Post	Perf. $\times$ Post	Team Perf.
Coefficient	0.268***	-0.757	0.0218	0.126***
Std. Error	(0.0552)	(0.792)	(0.0789)	(0.0361)
Observations	540			
Pseudo R^2	0.410			
Year FE	Included			
Standard errors in parentheses				
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$				

Table 2 reports the baseline probit regression results for promotion to Formula One. The key variable of interest, the performance index, enters the model positively and is highly statistically significant ($p < 0.01$), indicating that stronger performance is associated with a higher probability of promotion. The coefficient on the performance index is economically meaningful, suggesting that performance signals play a central role in determining progression outcomes at the highest level of the junior formula system.

Table 3: Average Marginal Effects

	b	se	p
Performance Index	0.0246***	(0.0037)	0.000
Team Performance	0.0112***	(0.0032)	0.000
<i>N</i>	540		
Standard errors in parentheses			
* <i>p</i> < 0.10, ** <i>p</i> < 0.05, *** <i>p</i> < 0.01			

However, due to the nonlinear nature of the probit model, coefficient magnitudes are not directly interpretable in probability terms. To address this, Table 3 reports the average marginal effects (AMEs). The estimated AME for the performance index is approximately 0.0246, implying that a one-unit increase in the index, equivalent to a one standard deviation improvement in performance, is associated with an increase in promotion probability of roughly 2.46 percentage points. Given the low baseline promotion rate in the sample, this represents a substantial effect, reinforcing the importance of performance in selection decisions.

The control variable capturing team quality, proxied by the reversed constructors' championship position, is also positive and statistically significant. The marginal effect suggests that drivers competing for stronger teams have a higher likelihood of promotion, consistent

with the idea that team environment and visibility may influence selection decisions. Importantly, the inclusion of this control does not attenuate the effect of performance, indicating that individual performance remains a robust predictor even after accounting for team-level factors.

5.2 Institutional Change and Interaction Effects

To examine whether the effectiveness of performance signals changed following the formalization of the junior formula system, the model includes both a post-formalization dummy and an interaction term between the performance index and the post-formalization indicator. The results show that neither the post-formalization dummy nor the interaction term is statistically significant.

Table 4: Average Marginal Effects by Period

Variable	Pre-Formalization	Post-Formalization
Performance Index	0.0244*** (0.0047)	0.0248*** (0.0053)
Team Performance	0.0115*** (0.0031)	0.0108** (0.0035)
Observations	540	540
Standard errors in parentheses		
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$		

This suggests that, while the structure of the junior formula system evolved over time, the relationship between performance and promotion remained largely unchanged. Table 4 presents the marginal effects evaluated separately for the pre- and post-formalization periods. The estimated marginal effects of the performance index are highly similar across both periods, with values of approximately 0.0244 and 0.0248, respectively. The overlap in confidence intervals further indicates that there is no statistically meaningful difference between these effects.

Taken together, these findings imply that performance signals have remained consistently important in determining promotion outcomes, and that institutional changes did not significantly alter the underlying selection mechanism.

5.3 Robustness Checks

To ensure that the baseline results are not driven by specific modeling choices or data construction decisions, a range of robustness checks were conducted. The results from these alternative specifications are summarized in A.1.

First, the probit model was re-estimated using a logit specification. The results remain qualitatively unchanged, with the performance index retaining a positive and statistically

significant effect on promotion. This confirms that the findings are not sensitive to the choice of link function.

Second, the performance index was reconstructed using principal component analysis (PCA) to address potential concerns regarding the weighting of individual performance components. The PCA-based index yields consistent results, with a strong and significant relationship between performance and promotion, suggesting that the findings are not dependent on the specific construction of the index.

Third, the sample was restricted by trimming extreme values of the performance index to mitigate the influence of potential outliers. The results remain stable under this restriction, indicating that the observed relationship is not driven by a small number of high-performing individuals.

Fourth, the analysis was conducted separately for the pre- and post-formalization subsamples. In both cases, the performance index remains a significant predictor of promotion, providing further evidence that the main findings are not driven by temporal variation in the sample.

Finally, alternative variance estimators were employed, including both clustered and heteroskedasticity-robust standard errors. The statistical significance of the key variables remains unchanged, indicating that the results are not sensitive to assumptions regarding the error structure.

5.4 Evaluation and Interpretation

Overall, the results provide strong evidence that performance is a key determinant of promotion to Formula One. The consistency of the performance effect across multiple specifications, model forms, and subsamples suggests that this relationship is both robust and economically meaningful. At the same time, the absence of a significant interaction effect indicates that institutional changes in the junior formula system did not materially alter the importance of performance signals.

findings are broadly consistent with the signaling framework, where observable performance serves as a primary mechanism through which drivers demonstrate their ability to compete at higher levels. However, the persistence of the performance effect across institutional regimes suggests that selection criteria may be more stable than initially hypothesized.

5.5 Comparison with Lower-Tier Promotions

To provide a benchmark, the analysis is replicated using promotion outcomes from GP3/F3 to GP2/F2. In this lower-tier sample, performance remains a statistically significant predictor of promotion; however, the estimated effect is smaller than in the F2 to F1

transition. This is consistent with a less selective promotion process at lower levels, where opportunities are more abundant and constraints are weaker. The contrast suggests that while performance signals matter throughout the junior system, they play a more decisive role at the final step into Formula One, where competition for limited seats is most intense. The results from this regression are found in the C.2.

6 Conclusion

This paper examined whether the effectiveness of early-career performance signals in determining promotion to Formula One has changed following the formalization of the junior formula system. Using a probit framework applied to an unbalanced panel of junior drivers, I constructed a composite performance index capturing multiple dimensions of driver ability and estimated its effect on promotion outcomes. The results show that performance is a strong and statistically significant predictor of promotion. However, there is no evidence that the effectiveness of these signals has changed over time, as the interaction between performance and the post-formalization period is statistically insignificant and marginal effects remain similar across regimes.

These findings contribute to the literature on signaling and promotion in tournament-style labor markets by providing evidence that the informativeness of performance-based signals remains stable despite increased institutional structure. Rather than reflecting a weakening of meritocratic selection, the results suggest that observed changes in promotion outcomes are better understood as a function of structural constraints, highlighting the importance of distinguishing between signal effectiveness and opportunity.

From my perspective as an observer of the sport over the past decade, promotion into Formula One appears to have become less responsive to performance. The empirical results, however, suggest that this perception is not driven by a weakening of performance signals, but by reduced opportunities. The reduction in grid size, from 12 teams to 10 (until the planned expansion in 2026), and increasing driver longevity have limited turnover at the top level. As highlighted in the literature on “mediocre talent” by Peeters et al. (2022), incumbency and contract stability can allow mid-performing drivers to retain seats for extended periods, restricting entry for new drivers.

Despite the robustness of the findings, several limitations should be acknowledged. The analysis relies on observable performance metrics and does not capture other determinants of promotion, such as financial backing, driver academies, or team relationships. The construction of the performance index also imposes implicit weighting choices, and the relatively small number of promotions introduces concerns related to missing observations and limits statistical power.

With more time and data, the analysis could be extended using richer datasets incorporating contracts, sponsorships, and academy affiliations. Exploiting exogenous variation,

such as changes in team entry or exit, could help isolate the causal impact of structural constraints. Expanding the analysis to other motorsport series or comparable labor markets may also provide broader insights.

Overall, this paper shows that performance remains a central determinant of advancement to Formula One, and that its effectiveness has not diminished following institutional formalization. The perceived change in promotion dynamics is more likely driven by reduced capacity and lower turnover rather than a shift away from performance-based selection.

References

- Altonji, J. G. and Pierret, C. R. (2001). Employer learning and statistical discrimination. *Quarterly Journal of Economics* 116, no. 1, 313–350. <https://doi.org/10.1162/003355301556329>.
- Arcidiacono, P., Bayer, P., and Hizmo, A. (2010). Beyond signaling and human capital: Education and the revelation of ability. *American Economic Journal: Applied Economics* 2, no. 4, 76–104. <https://doi.org/10.1257/app.2.4.76>.
- FIA Formula 2 Championship (2024). *Standings Formula 2 2024*. <https://www.fiaformula2.com/Standings/Driver?seasonId=181>.
- FIA Formula 2 Championship (2025). *Standings Formula 2 2025*. <https://www.fiaformula2.com/Standings/Driver?seasonId=182>. Additional seasons (2017–2023) were obtained from the same official source using consistent data structures.
- FIA Formula 3 Championship (2024). *Standings Formula 3 2024*. <https://www.fiaformula3.com/Standings/Driver?seasonId=181>.
- FIA Formula 3 Championship (2025). *Standings Formula 3 2025*. <https://www.fiaformula3.com/Standings/Driver?seasonId=182>. Additional seasons (2019–2023) were obtained from the same official source using consistent data structures.
- Müller-Kock, A. and Budzinski, O. (2016). Market power and media revenue allocation in professional sports: The case of formula one. *Ilmenau Economics Discussion Papers*, vol. 20, no. 102. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2801836.
- Peeters, T., Szymanski, S., and Terviö, M. (2022). The survival of mediocre superstars in the labor market. *Journal of Law, Economics, and Organization* 38, no. 3, 840–888. <https://doi.org/10.1093/jleo/ewab035>.
- Rosen, S. and Sanderson, A. (2000). Labor markets in professional sports. *NBER Working Paper no. 7573*. <https://doi.org/10.3386/w7573>.
- Scarfe, R., Singleton, C., and Telemo, P. (2021). Extreme wages, performance, and superstars in a market for footballers. *Industrial Relations: A Journal of Economy and Society* 60, no. 1, 84–118. <https://doi.org/10.1111/irel.12270>.
- Wikipedia (2025a). *2015 GP3 Series*. https://en.wikipedia.org/wiki/2015_GP3_Series.
- Wikipedia (2025b). *2016 GP3 Series*. https://en.wikipedia.org/wiki/2016_GP3_Series. Additional seasons (2010–2014) were obtained from the same official source using consistent data structures.
- Wikipedia (2026a). *2015 GP2 Series*. https://en.wikipedia.org/wiki/2015_GP2_Series.

Wikipedia (2026b). *2016 GP2 Series*. https://en.wikipedia.org/wiki/2016_GP2_Series
 Additional seasons (2008–2014) were obtained from the same official source using consistent data structures.

Wikipedia (2026c). *2024 Formula 2 Championship*.
https://en.wikipedia.org/wiki/2024_Formula_2_Championship.

Wikipedia (2026d). *2024 Formula 3 Championship*.
https://en.wikipedia.org/wiki/2024_FIA_Formula_3_Championship.

Wikipedia (2026e). *2025 Formula 2 Championship*.
https://en.wikipedia.org/wiki/2025_Formula_2_Championship
 Additional seasons (2017–2023) were obtained from the same official source using consistent data structures.

Wikipedia (2026f). *2025 Formula 3 Championship*.
https://en.wikipedia.org/wiki/2025_FIA_Formula_3_Championship
 Additional seasons (2019–2023) were obtained from the same official source using consistent data structures.

A Appendix A: Final Model

A.1 Robustness and Summary Stats

Table A.1: Robustness Checks

	Baseline	Logit	PCA	Trimmed	Pre	Post
Performance	0.268*** (0.055)	0.498*** (0.107)	0.584*** (0.118)	0.198*** (0.072)	0.291*** (0.055)	0.257*** (0.066)
Post-Formalization	-0.757 (0.792)	-1.268 (1.641)	-0.611 (0.799)	0.051 (1.215)	–	–
Performance $\times$ Post	0.022 (0.079)	-0.003 (0.147)	0.029 (0.165)	0.003 (0.098)	–	–
Team Performance	0.126*** (0.036)	0.254*** (0.076)	0.109*** (0.037)	0.163* (0.101)	0.082** (0.039)	0.222*** (0.070)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	540	540	540	79	292	248
Pseudo R^2	0.410	0.402	0.419	0.435	0.400	0.434
Standard errors in parentheses $*p < 0.10$, $**p < 0.05$, $***p < 0.01$						

Table A.2: Full Summary Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
Championship Position	540	15.71	9.01	1	35
Year	540	2016.14	5.21	2008	2025
Points	540	53.26	66.14	0	341.5
Sprint Wins	248	0.49	0.85	0	4
Feature Wins	248	0.40	0.87	0	5
Podiums	540	2.26	2.92	0	16
Points Finishes	540	7.12	6.16	0	25
Total Wins	540	0.76	1.29	0	7
Promoted	540	0.080	0.271	0	1
Constructors Rank (GP2/F2)	248	6.38	3.10	1	11
Number of Races	540	17.36	7.92	1	28
Pole Positions	540	0.38	0.95	0	8
DNS/DNF/DSQ	540	3.22	2.19	0	12
DNF Rate	540	0.201	0.150	0	1
Win Rate	540	0.035	0.060	0	0.333
Pole Rate	540	0.018	0.045	0	0.381
Podium Rate	540	0.107	0.133	0	0.762
Points per Race	540	2.48	2.85	0	16.26
Reversed Champ Position	540	-15.71	9.01	-35	-1
Reversed Constructor Rank	540	-6.97	3.47	-13	-1
Constructors Rank (Alt)	292	7.48	3.68	1	13
Points Finish Rate	540	0.355	0.268	0	1.333
Post Formalization	540	0.459	0.499	0	1
$z(\text{Championship})$	540	0.000	1.000	-2.14	1.63
$z(\text{Points})$	540	0.000	1.000	-1.32	3.65
$z(\text{Pole Rate})$	540	0.000	1.000	-0.39	8.00
$z(\text{DNF Rate})$	540	0.000	1.000	-1.34	5.33
Performance Index	540	0.000	2.98	-8.07	11.58
Driver ID	540	132.28	74.79	1	259
Team ID	540	23.46	15.33	1	56
Constructors Rank	540	6.97	3.47	1	13

A.2 Variable Definitions and Construction

Table A.3: Variable Definitions

Variable	Definition
Promoted	Indicator equal to 1 if driver enters Formula One within three years of their final F2/GP2 season, and 0 otherwise.
Performance Index	Composite measure constructed as the sum of standardized performance metrics, capturing overall driver ability.
Reversed Championship Position	Championship finishing position reversed so that higher values indicate better performance.
Points Finish Rate	Number of points-scoring finishes divided by total races competed.
Pole Rate	Number of pole positions divided by total races competed.
DNF Rate	Share of races in which the driver did not finish, including DNS, DNF, and DSQ outcomes.
Team Performance	Constructors' championship position reversed so that higher values correspond to stronger teams.
Post-Formalization	Indicator equal to 1 for seasons from 2017 onward, and 0 otherwise.

$$\text{PerfIndex} = z(\text{Championship}) + z(\text{Points}) + z(\text{Pole}) - z(\text{DNF})$$

Table A.4: Correlation Matrix of Performance Variables

	Champ	Points	Pole	DNF
Championship	1.000			
Points Finish Rate	0.856***	1.000		
Pole Rate	0.485***	0.452***	1.000	
DNF Rate	-0.253***	-0.271***	-0.114***	1.000

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

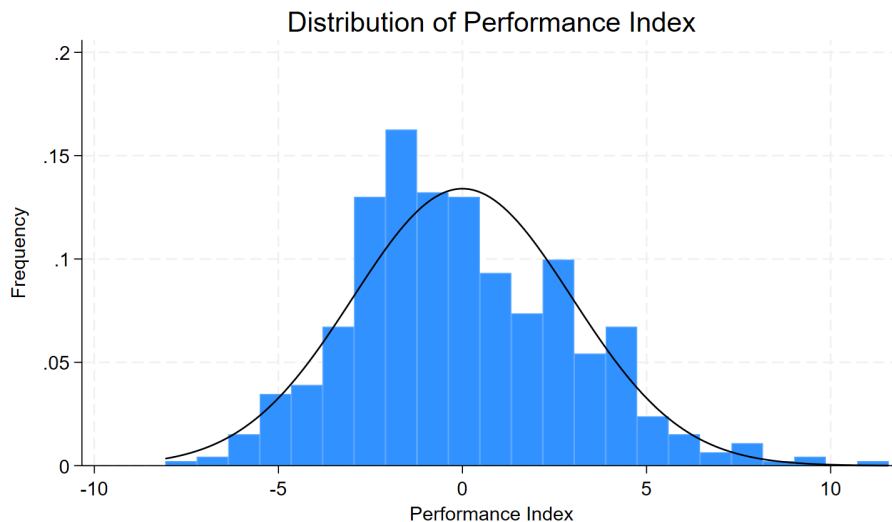


Figure A.1: Distribution of Performance Index

B Appendix B: Intermediate Model

Table B.1: Probit Regression with Disaggregated Performance Measures

	Coefficient	Std. Error
Reversed Championship Position	0.099**	(0.043)
Post-Formalization	-0.040	(0.896)
Performance $\times$ Post	0.074	(0.055)
Points Finish Rate	0.984	(1.233)
Pole Rate	2.831	(1.758)
DNF Rate	0.061	(1.530)
Team Performance	0.063	(0.045)
Year FE	Included	
Observations	540	
Pseudo R^2	0.444	
Standard errors clustered at the driver level		
Note: Separation present in sample		
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$		

Table B.2: Firth Logit Regression (Penalized Likelihood)

	Coefficient	Std. Error
Reversed Championship Position	0.167**	(0.082)
Post-Formalization	-0.145	(1.357)
Performance $\times$ Post	0.061	(0.103)
Points Finish Rate	2.472	(2.127)
Pole Rate	3.614	(2.880)
DNF Rate	1.543	(2.906)
Team Performance	0.093	(0.101)
Year FE	Included	
Observations	540	
Penalized likelihood estimation (Firth logit)		
Addresses separation in rare-event setting		
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$		

C Appendix C: Control Data (GP3/F3 - GP2/F2)

C.1 Summary Stats

Table C.1: Summary Statistics (F3/GP3 Sample)

Variable	Obs	Mean	Std. Dev.	Min	Max
<i>Performance Outcomes</i>					
Championship Position	480	17.794	10.153	1	40
Reversed Championship Position	480	-17.794	10.153	-40	-1
Points	480	38.729	51.783	0	253
Total Wins	480	0.471	0.918	0	5
Podiums	480	1.469	2.178	0	14
Points Finishes	480	5.027	4.894	0	17
Pole Positions	480	0.258	0.699	0	5
DNF/DNS/DSQ	480	2.152	1.728	0	9
Number of Races	480	14.292	5.674	2	20
<i>Rate Variables</i>					
Points Finish Rate	480	0.313	0.288	0	1
Pole Rate	480	0.015	0.042	0	0.278
Podium Rate	480	0.088	0.129	0	0.824
Win Rate	480	0.028	0.055	0	0.313
DNF Rate	480	0.156	0.134	0	1
Points per Race	480	2.330	2.974	0	14.882
<i>Constructed Variables</i>					
Performance Index	480	0.000	2.542	-3.645	10.062
Team Performance	480	-5.492	2.775	-10	-1
Post-Formalization	480	0.521	0.500	0	1
<i>Outcomes</i>					
Promotion to F2	480	0.192	0.394	0	1
Promotion to F1	480	0.010	0.102	0	1
<i>Other</i>					
Year of Season	480	2017.625	5.007	2010	2025
Constructors' Position	480	5.492	2.775	1	10

C.2 Baseline Regression

Table C.2: Promotion to F2/GP2 (Control Sample)

	Coefficient	Std. Error
Performance Index	0.294***	(0.043)
Post-Formalization	0.743*	(0.446)
Performance $\times$ Post	0.066	(0.061)
Team Performance	0.021	(0.034)
Year FE	Included	
Observations	480	
Pseudo R^2	0.323	
Standard errors clustered at the driver level		
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$		

C.3 Promotion Patterns

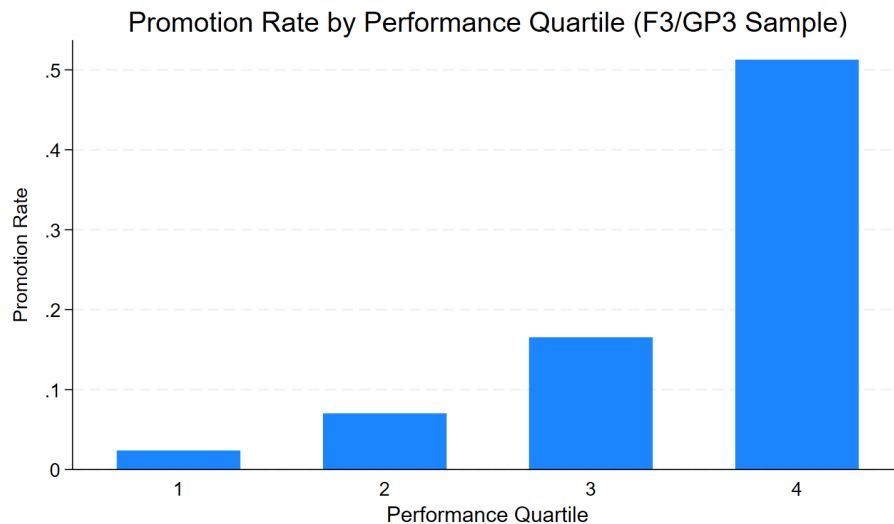


Figure C.1: Promotion Rate to GP2/F2 by GP3/F3 Performance Quartile

Figure C1 illustrates the relationship between performance and promotion in the F3/GP3 sample. Promotion rates increase steadily across performance quartiles, indicating a strong and monotonic relationship between performance and advancement. This pattern contrasts with the more discontinuous promotion dynamics observed in the F2-to-F1 transition, consistent with a less capacity-constrained environment.

C.4 Rare Case: GP3 to F1 Direct Promotion

Table C.3: Direct Promotion to F1 (Control Sample)

	Coefficient	Std. Error
Performance Index	0.229***	(0.073)
Team Performance	0.084	(0.092)
Year FE	Included	
Observations	136	
Pseudo R^2	0.291	
Standard errors clustered at the driver level		
Post-formalization terms omitted due to insufficient variation		
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$		

Performance remains a statistically significant predictor of direct promotion to Formula One, even within this smaller subsample. However, such transitions are extremely rare, as reflected in both the low number of observations and the limited variation in institutional variables, which prevents the estimation of interaction effects. Direct promotion to F1 is therefore less likely to follow a systematic, performance-driven pathway and may instead reflect exceptional circumstances. These include unique driver attributes, timing, or pre-existing relationships with teams and driver academies that are not captured by observable performance metrics. As a result, while performance remains relevant, these estimates should be interpreted as suggestive rather than definitive, reflecting a more idiosyncratic promotion process.